MATH 111 Calculus 1

Exponential functions

and their derivatives

Week 5 · Lecture 12 · §6.2

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

While you wait

Sit with your team. $2^{1/2} = \sqrt{2}$. Write $2^{3/2}$ and $2^{-1/2}$ with roots. Then: what could $2^{\sqrt{2}}$ mean, when $\sqrt{2}$ is not a fraction?

STAND UP

"The derivative of 2^x is $x \cdot 2^{x-1}$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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Verdict

FALSE. Check at $x = -1$: the formula gives $-1 \cdot 2^{-2} = -\frac{1}{4} < 0$. But 2^x increases, so every difference quotient $\dfrac{2^{-1+h} - 2^{-1}}{h}$ is positive, and a limit of positive numbers is not negative. The power rule is for x^n: the base varies, the exponent is fixed. In 2^x the *exponent* varies.

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Integers: $2^n = 2 \cdot 2 \cdots 2$ (n factors), $2^0 = 1$, $2^{-n} = \dfrac{1}{2^n}$.

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But $\sqrt{2}$ is not a fraction. Squeeze it between fractions — 2^r increases with r:

$$1.4 < \sqrt{2} < 1.5 \qquad 2.6390 < 2^{\sqrt{2}} < 2.8285$$
$$1.41 < \sqrt{2} < 1.42 \qquad 2.6573 < 2^{\sqrt{2}} < 2.6759$$
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Over *all* fractions $r < \sqrt{2} < s$, exactly one number lies above every 2^r and below every 2^s (a property of the real numbers: frame 4); that number *is* $2^{\sqrt{2}} = 2.66514\ldots$

Exponential functions

Definition

Let $b > 0$. For every real x: $b^x = \lim_{\substack{r \to x \\ r \text{ rational}}} b^r$. $f(x) = b^x$ is the exponential function with base b.

The limit is the ε–δ limit of Lecture 3, with only rational $r \neq x$ allowed.

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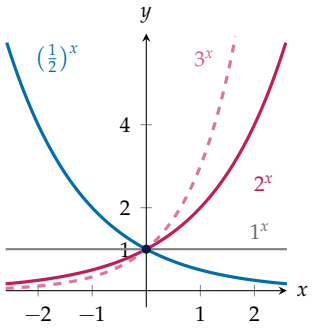
Taken without proof

The limit exists for every real x (for rational x it is the school value); b^x is continuous; for $b > 1$, b^x is increasing, and $b^x \to \infty$ as $x \to \infty$; and the laws of exponents hold for all real x, y and all $a, b > 0$: $b^{x+y} = b^x\, b^y$, $b^{x-y} = \dfrac{b^x}{b^y}$, $\left(b^x\right)^y = b^{xy}$, $(ab)^x = a^x\, b^x$.

The proofs need the construction of the real numbers; for rational exponents you know these facts from school.

Three kinds of exponential functions

$0 < b < 1$: decreasing; $b = 1$: the constant 1; $b > 1$: increasing. All pass through $(0, 1)$: $b^0 = 1$.

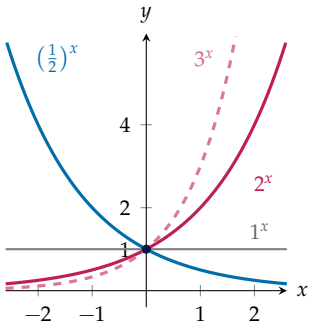


2^x and $(\frac{1}{2})^x$: mirror images

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$\left(\frac{1}{b}\right)^x = b^{-x}$: the graph of b^x, reflected in the y-axis.



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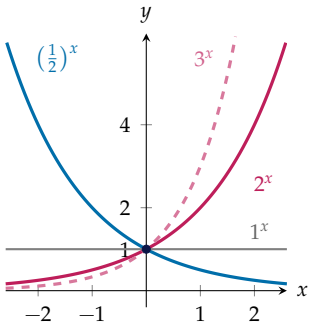
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For $b > 0$, $b \neq 1$

Domain $\mathbb{R}$; $b^x > 0$ for every x; range $(0, \infty)$.

If $b > 1$: $\lim_{x \to \infty} b^x = \infty$, $\lim_{x \to -\infty} b^x = 0$.

If $0 < b < 1$: $\lim_{x \to \infty} b^x = 0$, $\lim_{x \to -\infty} b^x = \infty$.



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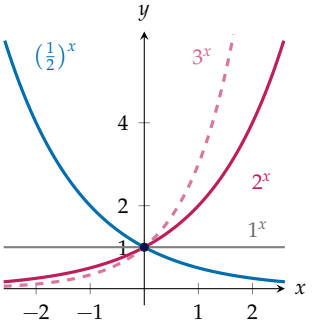
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If $0 < b < 1$: $\lim_{x \to \infty} b^x = 0$, $\lim_{x \to -\infty} b^x = \infty$.

The x-axis is a horizontal asymptote.

From frame 4: $b^x\, b^{-x} = 1$ and $b^x = \left(b^{x/2}\right)^2$, so $b^x > 0$;
$b^{-x} = 1/b^x$ gives the limit at $-\infty$ and the case $0 < b < 1$;
the range: the limits and the IVT (Lecture 5).



2^x and $\left(\frac{1}{2}\right)^x$: mirror images

Question 33

$$\lim_{x \to -\infty} \frac{5 \cdot 2^x + 3}{2^x + 1} = \ ?$$

A. 3

B. 5

C. 4

D. The limit does not exist.

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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THINK & TALK · 1 minute

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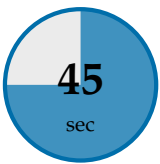
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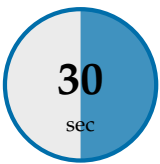
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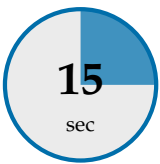
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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

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Hands down. Thank you.

A B C D

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EXPLAIN

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Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

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Answer

A. As $x \to -\infty$, $2^x \to 0$ (base $2 > 1$), so the quotient tends to $\dfrac{0+3}{0+1} = 3$. B is the limit at $+\infty$ — divide by 2^x: $\dfrac{5 + 3 \cdot 2^{-x}}{1 + 2^{-x}} \to 5$; the direction matters, as in Question 25. C is the value at $x = 0$, $\frac{8}{2} = 4$: a limit at $-\infty$ is not a value. D: dividing by 2^x here gives "$\frac{\infty}{\infty}$", read as "does not exist"; but the top tends to 3 and the bottom to 1.

The derivative of b^x

Theorem

Let $b > 0$ and $f(x) = b^x$. If f is differentiable at 0, then f is differentiable at every x, and $f'(x) = f'(0)\, b^x$.

One more fact taken without proof: b^x *is* differentiable at 0.

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$= b^x \lim_{h\to 0} \dfrac{b^h - 1}{h} = b^x f'(0)$ $\quad b^x$ does not depend on h $\square$

The last limit exists ($b^0 = 1$, so it is $f'(0)$); hence so does each limit before it: f is differentiable at x.

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The slope is proportional to the height

At every point, the slope of b^x is the height b^x times one fixed number: the slope at 0.

How big is the slope at 0?

The slope at 0 is $f'(0) = \lim_{h \to 0} \dfrac{b^h - 1}{h}$. For $b = 2$ and $b = 3$, from both sides (rounded to 5 decimals):

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h	$\dfrac{2^h - 1}{h}$	$\dfrac{3^h - 1}{h}$	h	$\dfrac{2^h - 1}{h}$	$\dfrac{3^h - 1}{h}$
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So $(2^x)' \approx 0.693 \cdot 2^x$ and $(3^x)' \approx 1.099 \cdot 3^x$. The slope at 0 is below 1 for base 2 and above 1 for base 3.

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Somewhere between 2 and 3 there should be a base whose slope at 0 is exactly 1.

The number e

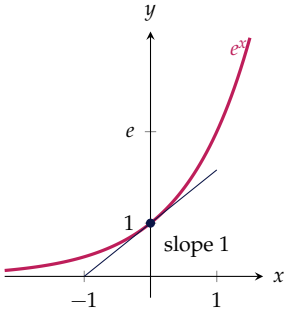
Definition

e is the base with slope 1 at 0: $\lim_{h \to 0} \dfrac{e^h - 1}{h} = 1.$

e^x is the natural exponential function;

$e = 2.71828\ldots$

Such a base exists (taken without proof), and there is only one (Lecture 13).



tangents at $(0, 1)$ and at $(1, e)$

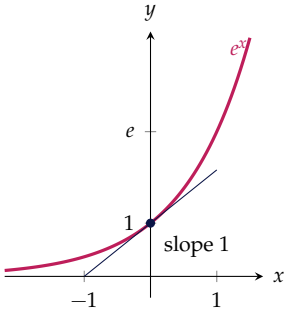
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$\dfrac{d}{dx} e^x = e^x$ — frame 9 with $f'(0) = 1$. At every point (x, e^x), the slope of the tangent is e^x.



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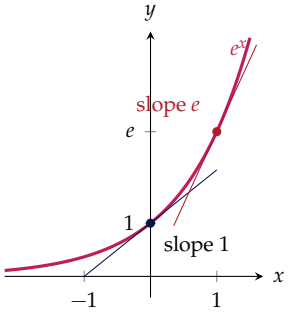
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At $(1, e)$, for instance, the slope is e.



tangents at $(0, 1)$ and at $(1, e)$

The chain rule with e^u

With the chain rule (g differentiable)

$$\frac{d}{dx}\, e^{g(x)} = \underbrace{\exp}_{\substack{\text{derivative}\\ \text{of outside}}} \underbrace{\big(g(x)\big)}_{\substack{\text{evaluated}\\ \text{at inside}}} \cdot \underbrace{g'(x)}_{\substack{\text{derivative}\\ \text{of inside}}} = e^{g(x)}\, g'(x)$$

$\exp u$ is e^u, written on one line: the outside is e^u, and its derivative is e^u again.

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(a) $y = e^{4x-x^2}$: $y' = e^{4x-x^2} \cdot (4-2x)$.

(b) $y = x^3 e^{-2x}$: product rule, then the chain:
$y' = 3x^2\,e^{-2x} + x^3\,e^{-2x} \cdot (-2) = x^2 e^{-2x}(3-2x)$.

Question 34

Let $f(x) = e^{1-x^2}$. What is $f'(1)$?

A. 1 **B.** -2

C. 2 **D.** 0

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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THINK & TALK · 1 minute

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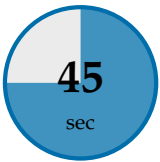
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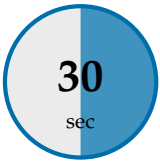
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VOTE

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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

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Hands down. Thank you.



Question 34

EXPLAIN

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Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

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Answer

B. $f'(x) = e^{1-x^2} \cdot (-2x)$, so $f'(1) = e^0 \cdot (-2) = -2$. A is e^{1-x^2} at 1: the inside's derivative forgotten (Lecture 10's mistake, again). C loses the minus sign of $-x^2$. D differentiates the number $f(1) = e^0 = 1$ — evaluating first and differentiating second is the wrong order — or uses the power rule on e^u.

Question 35

Find y' if $y = e^{-4x}\sin 5x$.

A. $e^{-4x}\,(5\cos 5x - 4\sin 5x)$

B. $e^{-4x}\,(5\cos 5x + 4\sin 5x)$

C. $e^{-4x}\,(\cos 5x + \sin 5x)$

D. $-20\,e^{-4x}\cos 5x$

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60 sec

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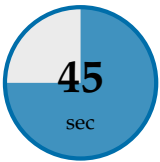
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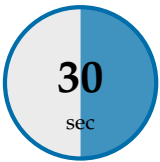
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0
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A B C D

Question 35

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Answer

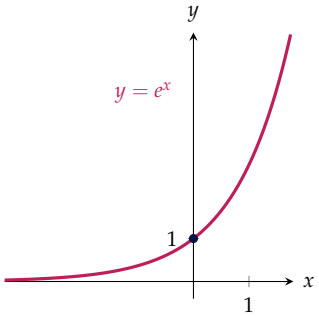
A. The product rule, and the chain rule in each factor:
$y' = e^{-4x}(\cos 5x)(5) + (\sin 5x)\,e^{-4x}(-4) = e^{-4x}(5\cos 5x - 4\sin 5x)$. B loses the minus sign of the inside $-4x$. C forgets both inside derivatives, 5 and -4. D differentiates each factor and multiplies the results: $(fg)'$ is not $f'g'$.

Properties of the natural exponential function

$f(x) = e^x$ is an increasing continuous function with domain $\mathbb{R}$ and range $(0, \infty)$. Thus $e^x > 0$ for all x. Also

$$\lim_{x \to -\infty} e^x = 0 \qquad \lim_{x \to \infty} e^x = \infty$$

So the x-axis is a horizontal asymptote of $f(x) = e^x$.



the natural exponential function

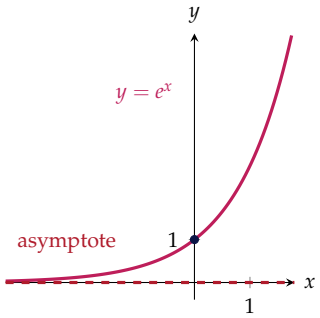
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Nothing new: this is frames 4 and 5 for the base $b = e$ ($e = 2.718\ldots > 1$).



the natural exponential function

A limit with e^x

Example. Find $\lim_{x \to \infty} \frac{e^{2x}}{e^{2x} + 1}$.

A limit with e^x

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We have used the fact that $t = -2x \to -\infty$ as $x \to \infty$, and so

$$\lim_{x\to\infty} e^{-2x} = \lim_{t\to-\infty} e^{t} = 0.$$

Why this is allowed: if $e^t < \varepsilon$ for all $t < -T$, then $e^{-2x} < \varepsilon$ for all $x > T/2$.

A little history · 1690–1748

1690. **Jacob Bernoulli**, in a paper on interest: one coin at 100% a year, with the interest added n times a year, becomes $\left(1 + \frac{1}{n}\right)^n$ coins. What happens as n grows? He showed that the limit lies between $2\frac{1}{2}$ and 3.

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1748. In the *Introductio*, Euler writes $a^\omega = 1 + k\omega$ for an infinitely small ω — in today's words, k is the slope of a^x at $(0,1)$ — and takes the base with $k = 1$, which he prints as 2.71828182845904523536028. Frame 11's definition is his.

Today in two lines

1. b^x is defined for every real x, through rational exponents: continuous, positive, increasing for $b > 1$ and decreasing for $0 < b < 1$, with the laws of exponents; for $b \neq 1$ the x-axis is an asymptote.

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This week:

Homework: §6.2 — read and work out the examples. Next: logarithmic functions and their derivatives (§6.3, §6.4) — where the 0.693 . . . of frame 10 gets its name.

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: *"The derivative of* 2^x *is not* $x \cdot 2^{x-1}$ *because …" (one sentence, or one number).*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.